

Table of contents

Refraction for identifying matter (property of matter):

$$\begin{aligned}
 \eta &\stackrel{opt.}{=} \frac{\sin(\alpha)}{\sin(\beta)} = \frac{\eta_\beta}{\eta_\alpha} \stackrel{prism}{=} \frac{\sin[\frac{1}{2}(\varpi + \varsigma_{min})]}{\sin[\frac{1}{2}\varpi]} \\
 &\stackrel{phys.}{=} \sqrt{\epsilon_r \mu_r} = \frac{\lambda_\alpha}{\lambda_\beta} = \frac{v_{ph,\alpha}}{v_{ph,\beta}} = \frac{1}{2} \frac{\lambda}{d_l} \frac{n}{\cos(\beta)} = \sqrt{\left(\frac{n}{2} \frac{\lambda}{d_l}\right)^2 + \sin(\alpha)^2} \\
 &\stackrel{dispersion}{=} \eta(\lambda) = \eta(\omega) \stackrel{anisotropy}{=} \eta(r)
 \end{aligned} \tag{1}$$

Vacuum $\iff \eta = 1$: [1]

$$f = FocalLength \quad D = RefractivePower(dioptry) \tag{2}$$

$n \equiv$ order (“quantum”):

$$n = \frac{\nu}{c} \Delta s = \frac{\Delta \varphi}{2\pi} = 2\eta \frac{d_l}{\lambda} \sqrt{\eta^2 - \sin(\alpha)^2} = 2\eta \frac{d_l}{\lambda} \cos(\beta) = \frac{q}{e_0} \in \mathbb{N} \tag{3}$$

$$v^2 = \left(\frac{\nu \lambda}{\eta}\right)^2 = \left(\frac{\omega}{k\eta}\right)^2 \tag{4}$$

$$\left(\frac{\eta}{c}\right)^2 = \epsilon \mu \quad \epsilon = \epsilon_0 \epsilon_r \quad \mu = \mu_0 \mu_r \tag{5}$$

where:

$d_l :=$ distance length

$\alpha :=$ irradiation angle, incoming radiation direction, from source/sender (Einstrahlungswinkel zur Normalen auf Grenzfläche des Mediums)

$\beta :=$ refraction angle for bented transmitted radiation A_t orientation in matter (Durchstrahlungswinkel zur Normalen aus Grenzfläche ins Medium)

[1] Brechkraft